MODULE 30 · DOMAIN 5 - GOVERNANCE

Masking, licensing and problematic data

Keeping sensitive data out while the application still works, using only data you have the right to use, and cleaning bad text at its source.

SECTION 5 of 6 EXAM WEIGHT 8% MODULE 30 of 40

admin@sthithapragna.com

Fix the water at the reservoir, not at every tap. **Problems in the data are cheapest and safest to fix before the data reaches the model.**

Where masking can happen

WHERE	TOOL
At rest, in tables	Unity Catalog column masks and ABAC policies
In pipelines, before indexing	ai_mask on text columns
At request time	Unity AI Gateway PII redaction on input
On the way out	PII redaction or blocking on output

ON THE EXAM

Mask as early as the use case allows. Data masked before indexing can never be retrieved unmasked.

Masking that still meets the performance objective

BLOCKING

Safe, but the task stops

- Request refused if data is found
- Nothing sensitive leaks
- Summaries and answers fail
- Objective missed

MASKING

Safe, and the task continues

- Sensitive parts replaced with placeholders
- Sentence structure preserved
- Model still summarises or answers
- Objective met

THE TIEBREAKER

When a scenario needs both protection and a working application, masking is the answer. Blocking protects by stopping the work.

Keeping quality after masking

- Use **typed placeholders**, such as [PERSON] or [EMAIL], so meaning survives.
- Keep placeholders **consistent** within a document, so the model can follow who is who.
- Mask **before embedding**, so the index never contains the sensitive values.
- **Re-run the evaluation set** after masking, to prove the objective still holds.
- Over-masking removes the facts the task needs. Mask only what policy requires.

ON THE EXAM

Masking is a quality trade-off like any other. Measure it with the evaluation set, as module 20 taught.

Masking and filtering in Unity Catalog

CONTROL	WHAT IT DOES
Row filter	A function hides rows a user may not see
Column mask	A function replaces values a user may not see
ABAC policy	Applies filters and masks by governed tag, across whole catalogs or schemas
Data classification	Detects and tags sensitive data automatically

ON THE EXAM

ABAC policies, governed tags and data classification are generally available. Databricks recommends ABAC for rules across many tables.

Table-level masks, or ABAC policies?

TABLE-LEVEL

One table at a time

- Set on a specific table or column
- Managed by the table owner
- Fine for a few tables
- New tables need their own

ABAC POLICY

Rules by tag

- Attached at catalog or schema level
- Applies wherever a governed tag appears
- New tagged tables covered automatically
- Separates policy authors from data owners

THE TIEBREAKER

When the same protection must cover many tables, including future ones, use ABAC. For one or two tables, table-level masks are enough.

Having the right to use the data

- Check the licence permits **this use**: commercial, internal only, redistribution, derivatives.
- **Publicly readable is not free to use.** Websites have terms and copyright.
- Personal data needs a **lawful basis**, such as consent, for this purpose.
- Record source, licence and restrictions as **metadata or tags**, with lineage in Unity Catalog.
- Models have licences too, from modules 11 and 12.

> ON THE EXAM
>
> If the licence does not clearly permit the intended use, leave the source out until it does.

Publicly available, or licensed for this use?

PUBLICLY AVAILABLE

Anyone can read it

- On the open web
- Says nothing about reuse
- Terms may forbid scraping
- Copyright still applies

LICENSED FOR THIS USE

You may use it this way

- Terms cover your purpose
- Commercial or internal as needed
- Redistribution allowed if answers quote it
- Recorded with the source

THE TIEBREAKER

Paraphrasing or adding a disclaimer does not make unlicensed content licensed. Only the terms do.

When a source contains text you do not want repeated

- Toxic or inflammatory posts, biased material, outdated or confidential content.
- **Remove or replace** it at the source, before chunking and indexing.
- **Classify and filter** at scale, for example with ai_classify on a toxicity label.
- Choose a **cleaner or curated source** for the same knowledge.
- Keep an output guardrail as a **backstop**, not as the main fix.

> ON THE EXAM
>
> The guide asks for an alternative mitigation. The strongest answer changes the data, not the prompt.

Harbourline cleans its sources

01
Tag
Classification tags PII columns.

02
Protect
ABAC masks them for non-claims staff.

03
Mask
ai_mask on adjuster notes before indexing.

04
License
Bought market report: internal use only, excluded.

05
Clean
Abusive complaint text filtered out of the index.

ON THE EXAM

Every fix happened at the reservoir, before the model saw anything. The guardrails from module 17 remain as backstops.

> Support call transcripts are sent to an externally hosted model to produce summaries. Policy forbids customer names and phone numbers leaving the company.
>
> The summaries must still be useful to the support team.

Which approach meets both requirements?

A Block every transcript that contains any personal data

B Mask names and numbers with placeholders before the call

C Delete all transcripts that mention a customer by name

D Tell the model in its prompt to forget personal details

◆ PAUSE AND DECIDE

Which approach meets both requirements?

A Block transcripts

Nearly every transcript has a name. No summaries.

B Mask before the call

Nothing sensitive leaves, and summaries still work.

C Delete transcripts

Destroys the data the task needs.

D Prompt to forget

The data has already left by then.

THE SIGNAL

Protection plus a useful summary. Masking is the guardrail that meets the performance objective.

> A rag assistant indexes raw HR case notes, including employees' health details. Managers using it sometimes see those health details in answers.
>
> Managers still need the assistant for the rest of the case information.

What is the best fix?

A. Add a prompt instruction telling the model not to repeat health data

B. Enable the reranker so health details rank lower in the results

C. Show managers a warning banner before they use the assistant

D. Mask or remove health details before chunking and indexing

◆ PAUSE AND DECIDE

What is the best fix?

A Prompt instruction

The details are still retrieved. A prompt is not a control.

B Reranker

Reorders by relevance. Health details still appear.

C Warning banner

Warns about the leak without stopping it.

D Mask before indexing

The index never holds them, so no answer can reveal them.

THE SIGNAL

Sensitive values in the index itself. Fix it at the reservoir: mask before indexing.

> A team wants to add two sources to a customer-facing assistant: articles scraped from an industry news site, and a purchased report licensed for internal research only.
>
> Legal has not reviewed either source.

What should the team do?

A Exclude both until licences permit this use, and record the terms

B Include both, since the news content is publicly readable online

C Include both, but paraphrase answers so the wording is different

D Include both, and add a disclaimer to every generated answer

◆ PAUSE AND DECIDE

What should the team do?

A Exclude, record the terms

Neither is licensed for customer-facing use yet.

B Publicly readable

Readable is not reusable. Terms and copyright apply.

C Paraphrase

Rewording does not grant rights to the content.

D Disclaimer

A disclaimer does not create a licence.

THE SIGNAL

Internal research only, used in a customer-facing assistant, and nothing reviewed. Leave them out until the terms say yes.

> A product assistant retrieves from a community forum. Some posts are abusive and inflammatory, and the assistant occasionally echoes their language.
>
> The forum also holds much useful troubleshooting advice.

Which mitigation should the engineer recommend?

A. Raise the temperature so answers drift away from the posts

B. Use a larger context window so more calm posts fit alongside

C. Filter or replace the toxic posts in the source before indexing

D. Rely on the model's own safety training to ignore the posts

◆ PAUSE AND DECIDE

Which mitigation should the engineer recommend?

A. Higher temperature
More randomness, and just as likely to echo the posts.

B. Larger context
The toxic posts are still retrieved alongside.

C. Filter at the source
Keeps the useful advice, removes the toxic text.

D. Model safety training
Not reliable, and it failed already.

THE SIGNAL

Problematic text in the data source. The strongest alternative changes the data, and keeps what is useful.

What to carry forward

- Fix data problems **at the reservoir**, before the model sees anything.
- **Masking** protects and keeps the task working. Blocking protects by stopping it.
- Unity Catalog: row filters, column masks, **ABAC by governed tag**, data classification.
- **Publicly readable is not licensed.** Paraphrase and disclaimers do not fix it.
- Problematic text: filter or replace at source. Guardrails as backstop.

> ON THE EXAM
>
> That completes section 5. Next, section 6: module 31 covers evaluating with MLflow judges and scorers.